Thanumalayaperumal T

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.96
SR MATRIC HR SEC SCHOOL	Class XII (Tamil Nadu State Board of School Education (TNBSE))	2021	95.37%
SR MATRIC HR SEC SCHOOL	Class X (Tamil Nadu State Board of School Education (TNBSE))	2019	93.80%

SKILLS

Thin-Film Deposition Techniques - Fabrication of thin films using advanced physical vapour deposition techniques:

- Pulsed Laser Deposition (PLD)
- Electron-beam Evaporation (E-beam PVD)

Materials Synthesis and Processing

- Expertise in advanced materials synthesis techniques, including Co-precipitation, Sol Gel, Hydrothermal, Solvothermal

Instrumentation

- XRD (Powder and Thin Film)
- XPS (X-ray Photoelectron Spectroscopy)
- AFM (Atomic force microscopy)
- I-V Characteristics (RRAM Switching)

Simulation & Modelling

- Phase-Field Modelling, Multiphysics Simulation, Semiconductor Device Simulation, MOOSE Framework

Programming & Computational Tools

- Python, Julia, C++, Java, MATLAB, Jupyter Notebook, Linux/Ubuntu, Git, GitHub.

Software & Visualization

VESTA, Origin, ImageJ, Crystal Maker, Crystal Diffract

PROJECTS

Phase-Field Simulation of Conductive Filament Formation in RRAM

Jan 2026 - Apr 2026 | Supervisor: Prof. Saswata Bhattacharya | IIT Hyderabad | GitHub

- Developed a finite-difference phase-field model in Julia to simulate conductive filament formation and evolution in dielectric thin films for RRAM applications.
- Solved coupled Cahn-Hilliard and electrostatic equations to investigate electroforming dynamics, defect redistribution, and conductive-channel evolution.
- Analysed the influence of electric-field-driven defect migration on filament formation and resistive switching behaviour.

ML-Based Identification of Phantom Correlations in Materials Data

Jan 2026 - Apr 2026 | Supervisors: Prof. Saswata Bhattacharya, Prof. Anuj Goyal | IIT Hyderabad | GitHub

- Developed synthetic materials datasets to investigate spurious correlations and distinguish causal descriptors from correlated proxy features using machine learning.
- Applied correlation analysis, cross-family validation, permutation importance, and intervention-based validation to identify robust predictive features.
- Demonstrated limitations of conventional feature-selection methods and highlighted the role of causal inference in materials informatics.

Molecular Dynamics Simulation of Argon Melting Using Python

Jan 2026 - Apr 2026 | Supervisors: Prof. Saswata Bhattacharya, Prof. Anuj Goyal | IIT Hyderabad | GitHub

- Developed a molecular dynamics simulation engine from scratch in Python using the Lennard Jones potential and Velocity Verlet integration algorithm.
- Implemented periodic boundary conditions, Verlet neighbor lists, and structural analyses using radial distribution function (RDF) and mean square displacement (MSD).
- Investigated the solid-to-liquid phase transition of argon and estimated melting behavior consistent with reported literature values.

WORK & INTERNSHIP EXPERIENCE

Kerala State Electricity Board Limited (KSEB)

Intern

Jun 2023 | Thiruvananthapuram, Kerala

- Completed an internship at KSEB, gaining practical exposure to power-system operations, electrical distribution, and utility infrastructure.
- Gained insight into the operation and maintenance of electrical power systems in a utility environment.

RELEVANT COURSES

1. Semiconductor Devices.
2. Fundamentals of Semiconductor Materials.
3. Micro and Nanofabrication.
4. Electronic Packaging.
5. Imaging in Materials Science.
6. Vacuum Technology in Semiconductor.
7. Semiconductor Extraction, Purification and Growth.
8. Machine Learning and Data Analytics in Materials Science.
9. Electronic Structure and Atomistic Modeling of Materials.
10. Mesoscale Microstructure Modelling.
11. Semiconductor Devices Characterization Laboratory.
12. Semiconductor Materials Characterization Laboratory.
13. Attended various industry lectures by DRDO, Global Foundries, ISRO as part of MTech curriculum.

LEADERSHIP & PROFESSIONAL AFFILIATIONS

- Associate Professional Services Director, Rotaract Club of Chennai Capitals | 2023-2026

CERTIFICATIONS & FELLOWSHIPS

- Samsung Fellowship (Grade I) -- Samsung | May 2026
- ISWDP Advanced Level - TCAD, Process & Device Simulation - Samsung, Synopsys & IISc | Apr 2026
- ISWDP Level 2 - Advanced Device Modeling & Simulation - Samsung, Synopsys & IISc | Feb 2026
- ISWDP Level 1 - Semiconductor Device Technology & TCAD - Samsung, Synopsys & IISc | Feb 2026
- CeNSE Winter School on Semiconductor Technology - IISc Bengaluru | Dec 2025